
BY 513: Cellular Bioprocess Technology

Course Code	BY 513
Course Name	Cellular Bioprocess Technology
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	B. Tech. 3rd and 4th year, MS/M. Tech., Ph.D.
Distribution	Core for M. Tech. Biotechnology, elective for others
Approval	9th Senate (May 2016)

Course Contents

- **Introduction to Cell culture technology for Bioprocessing**

Cell culture engineering, cell culture products (metabolites, enzymes etc), Cellular systems as molecular factories – plants and microbial systems relevant to industries. Introduction to Bioprocessing. [3 Lectures]

- **Medium Design and kinetics of cell cultivation**

Optimization of cell growth environment. A guide for medium design. Types of media and classes of medium components. Components of different cell culture medium. Medium for the Industrial production culture, Stoichiometry and Kinetics of Cell Cultivation – composition, cell mass and size, Quantitative Description of Cell Growth & Product Formation, Kinetic Model of Cell Growth. Monod Model and its Derivatives. [14 Lectures]

- **Fermentation technology, Bioreactor design principles and operating mode**

Fermentation technology, Bioreactor types and design principles -Simple Stirred Tank Bioreactor, Airlift Bioreactor, Fluidized Bed Bioreactor, Membrane Bioreactor. Operating Mode of Bioreactors - Batch, fedbatch and Continuous Processes. Control strategies of physiological parameters - Oxygen transfer, redox, pH etc. Growth rates. Product recovery and quality analysis. Introduction to Analytical techniques, Scaling Up and Scaling Down for Cell Culture Bioreactors. [16 Lectures]

- **Metabolic engineering, CBP technology and modeling of cellular factories**

Developing optimal host cells by rational metabolic engineering – an overview. Modelling of metabolic pathways of cellular Systems. Consolidated Bioprocessing (CBP) technology in the context of biofuels and sustainable chemicals. [9 Lectures]

Text and Reference Books

1. Wei-Shou Hu, **Cell Culture Bioprocess Engineering**.
2. **Encyclopedia of Bioprocess Technology**, John Wiley and Sons.
3. Willem H. van Zyl, Riaan den Haan and Daniel C. la Grange, **Developing organisms for consolidated bioprocessing of Biomass to ethanol**, 2011.
4. Other relevant tutorial material and scientific publications will be provided.